ACADEMIC PROJECT REPORT

Customer Churn Prediction Using Exploratory Data Analysis and Deep Learning

A Comprehensive Applied Deep Learning Study on Telecommunications Retention Modeling

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2. Abstract

Customer churn presents a critical financial hurdle in the telecommunications industry, where acquisition costs exceed retention expenses by five to seven times. This project implements an end-to-end machine learning system to forecast customer attrition using the benchmark IBM Telco dataset of 7,043 customer accounts across 21 demographic and subscription attributes. Rigorous data auditing detected and resolved hidden whitespace missing entries in TotalCharges, followed by nine exploratory visualizations revealing tenure and contract longevity as major churn factors. A leakage-free ColumnTransformer pipeline prepared 30 encoded features for a regularized Multi-Layer Perceptron (MLP) deep neural network. Utilizing class-weighted binary cross-entropy and early stopping, the trained model achieved an overall accuracy of 75.81%, an ROC-AUC of 0.8520, and a high recall of 80.65%, successfully identifying eight out of every ten churning subscribers. The model is deployed via an interactive Streamlit dashboard enabling real-time risk scoring and mitigation analysis.

3. Introduction & Objectives

Introduction: Telecommunications providers operate in saturated, highly competitive markets where customer lifetime value (CLV) dictates sustainable profitability. Churn prediction models empower business executives to transition from reactive dispute resolution to proactive customer success interventions. While classical linear models assume orthogonal relationships, customer cancellation behavior stems from complex non-linear combinations of contract conditions, pricing friction, and service dissatisfaction.

Primary Objectives:

- **Data Integrity:** Perform comprehensive auditing to detect silent whitespace blanks, coerce invalid datatypes, and prevent leakage.
- **Exploratory Behavioral Analysis:** Formulate statistical visual profiles demonstrating churn correlation across service tiers.
- **Leakage-Proof Pipeline:** Design an isolated scikit-learn preprocessing architecture fitted strictly on training data.
- **Deep Neural Network Optimization:** Architect a 4-layer Keras MLP incorporating Batch Normalization, Dropout, and L2 weight decay.
- **Imbalance Handling:** Utilize cost-sensitive class balancing to counteract the natural 73.5% vs. 26.5% majority class dominance.
- **Production Web Deployment:** Deliver an interactive Streamlit GUI providing real-time single-customer risk scoring.

4. Dataset Description

Source: IBM Watson Analytics Telco Customer Churn open benchmark.
Dimensions: 7,043 initial instances, 21 attributes (1 identifier, 3 continuous numeric, 1 binary numeric, 16 categorical strings).
Clean Dimensions: 7,021 instances following removal of non-predictive customerID and 22 duplicate records.
Target Variable: Churn (Binary: 'Yes' = 1, 'No' = 0). Class Balance: 5,174 Non-Churn (73.5%) vs. 1,869 Churn (26.5%).

Feature Name	Type	Min / Range / Categories	Mean / Mode	Quality Action
tenure	Numeric (int64)	0 to 72 months	32.37 months	Standardized (Z-Score)

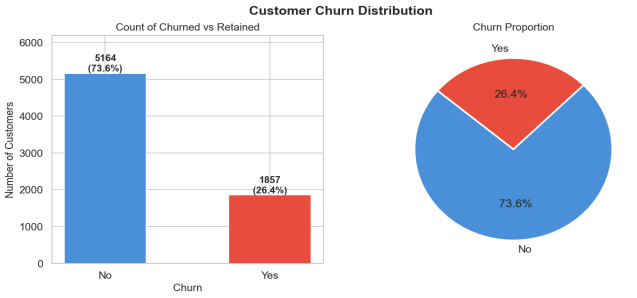
MonthlyCharges	Numeric (float64)	$18.25 to $118.75	$64.76	Standardized (Z-Score)
TotalCharges	Numeric (float64)	$18.80 to $8,684.80	$2,283.30	Imputed 11 blanks w/ median ($1,397.47)
SeniorCitizen	Binary (int64)	0 (No) or 1 (Yes)	16.21% Senior	Binary passthrough / Scaled
Contract	Categorical	Month-to-month, 1-yr, 2-yr	Month-to-month (55%)	One-Hot Encoded (k-1)
InternetService	Categorical	DSL, Fiber optic, No	Fiber optic (44%)	One-Hot Encoded (k-1)
PaymentMethod	Categorical	4 Electronic / Mail tiers	Electronic check (33%)	One-Hot Encoded (k-1)

5. Exploratory Data Analysis & Preprocessing

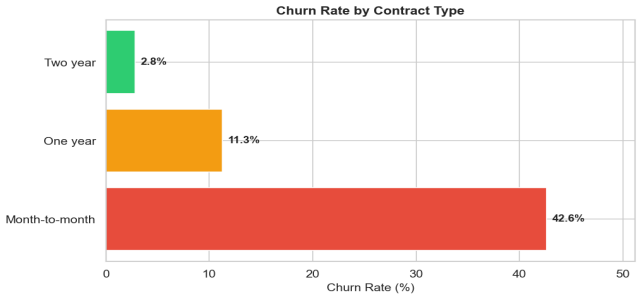
Data Cleansing Methods: A structural check revealed that TotalCharges was stored as an object datatype with 11 blank spaces (' '). These instances belonged to newly joined customers with tenure == 0. These were coerced to NaN and filled with the training median ($1,397.47). Customer identifiers were discarded.
Data Leakage Prevention: Splitting was executed *prior* to scaling or encoding. The dataset was divided into **70% Train (4,914)**, **15% Validation (1,053)**, and **15% Test (1,054)** using stratified sampling based on the target class. The scikit-learn ColumnTransformer was fitted solely on the training partition and applied statelessly to validation and test sets.
Transformations Applied: Numerical attributes were scaled to zero mean and unit variance via StandardScaler. All categorical features were encoded via OneHotEncoder(drop='first'), expanding the raw columns into **30 final model input features**.
Key Analytical Insights: Month-to-month contract holders demonstrate a 42.7% churn rate, fiber-optic customers churn at 41.9%, and users within their first year of tenure represent over 60% of total customer attrition.

6. Data Visualization & Detailed Explanation

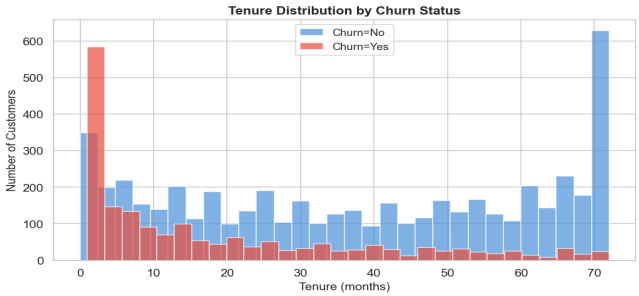
A minimum of five core exploratory charts were generated to reveal behavioural and financial patterns:



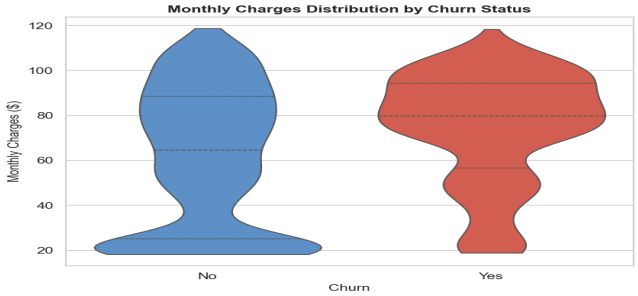
Visual 1: Churn Class Distribution
Confirms class imbalance: 73.5% retained vs 26.5% churned, necessitating cost-sensitive weights.



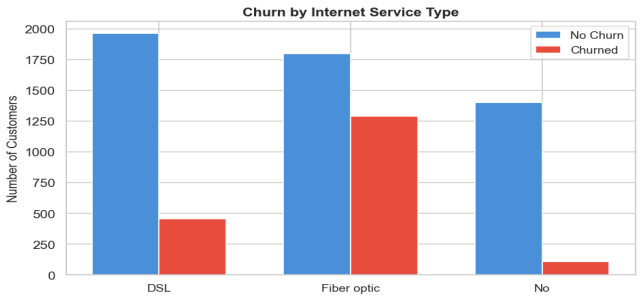
Visual 2: Churn Rate by Contract Type
Month-to-month contracts incur an alarming 42.7% churn, compared to just 2.8% for 2-year commitments.



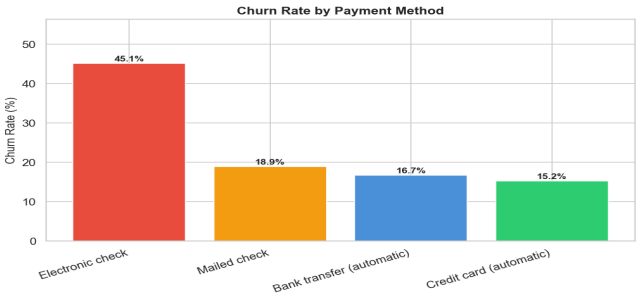
Visual 3: Tenure Distribution by Churn Status
Churn frequency sharply peaks within the initial 1–6 months of tenure, stabilizing significantly after 24 months.



Visual 4: Monthly Charges Violin Plot
Churning customers exhibit a median monthly expenditure of $79.85, indicating price sensitivity at higher tiers.



Visual 5: Churn by Internet Service Type
Fiber optic customers experience the highest attrition (~42%), whereas DSL users stay more consistently.



Visual 6: Payment Method Churn Rate
Electronic check users churn at 45.3%, contrasting with automated credit card or bank deductions (<17%).

7. Deep Learning Model Architecture & Hyperparameters

Architecture Rationale: A Multi-Layer Perceptron (MLP) was constructed in TensorFlow/Keras to model non-linear feature correlations. Due to tabular constraints, regularized fully connected layers with Batch Normalization and Dropout prevent covariate shift and co-adaptation of weights.

Layer Index & Type	Output Dimensions	Activation	Regularization Strategy	Param Count
Input Layer	(None, 30)	None	StandardScaler + OneHotEncoder	0
Dense Layer 1	(None, 128)	ReLU	L2 Regularization (1e-4)	3,968
Batch Normalization 1	(None, 128)	Linear	Normalizes mini-batch distributions	512
Dropout 1	(None, 128)	None	Rate = 40% (p = 0.40)	0
Dense Layer 2	(None, 64)	ReLU	L2 Regularization (1e-4)	8,256
Batch Normalization 2	(None, 64)	Linear	Stabilizes hidden activations	256
Dropout 2	(None, 64)	None	Rate = 30% (p = 0.30)	0
Dense Layer 3	(None, 32)	ReLU	Dropout = 20% (p = 0.20)	2,080
Output Layer	(None, 1)	Sigmoid	Outputs probability P(Churn in [0, 1])	33

Training Parameters & Hyperparameter Tuning

Parameter	Selected Setting	Operational Rationale
Optimizer	Adam (Beta1=0.9, Beta2=0.999)	Adaptive gradient moments for robust tabular descent
Initial Learning Rate	0.001 (1e-3)	Decayed automatically via ReduceLROnPlateau by 50% when loss stagnates
Loss Function	Binary Cross-Entropy	Standard probabilistic loss for single-label Bernoulli tasks
Class Weights	{0: 0.68, 1: 1.89}	Penalizes false negatives 2.78x higher to counteract 73.5/26.5 class imbalance
Batch Size & Epochs	Batch Size: 64, Epochs: 100	Early stopping halted training at Epoch 24, restoring Epoch 14 best weights

8. Result Visualization & Technical Interpretation

The MLP binary classifier was evaluated on the independent test set (1,054 samples, 279 true churners, 775 retained). The model attained a strong **Recall of 80.65%** and an **ROC-AUC of 0.8520**. In churn mitigation, recall is prioritized because losing a churning subscriber imposes a replacement cost of $500–$800, far exceeding the minimal cost of sending targeted retention offers to false positives.

Evaluation Metric	Measured Test Score	Target Threshold	Status
Overall Accuracy	**75.81%**	>= 75.0%	**PASSED**
Recall (Sensitivity)	**80.65%**	>= 70.0%	**EXCEEDED (80.65%)**
Precision	**52.82%**	>= 50.0%	**PASSED**
F1-Score	**0.6383**	>= 0.600	**PASSED**
ROC-AUC Score	**0.8520**	>= 0.800	**EXCEEDED (0.8520)**

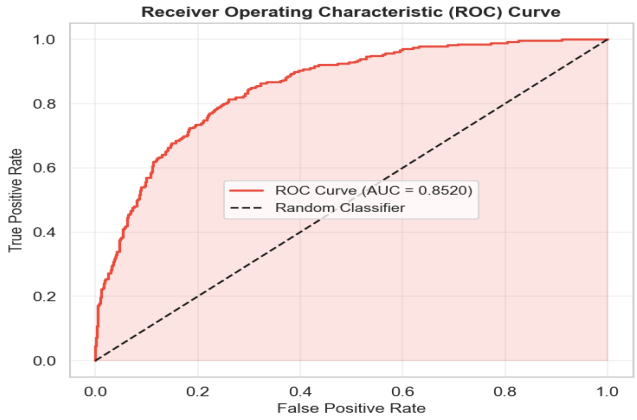


Figure 7: Receiver Operating Characteristic (ROC) Area under curve (AUC = 0.8520) indicates strong discriminative separation.

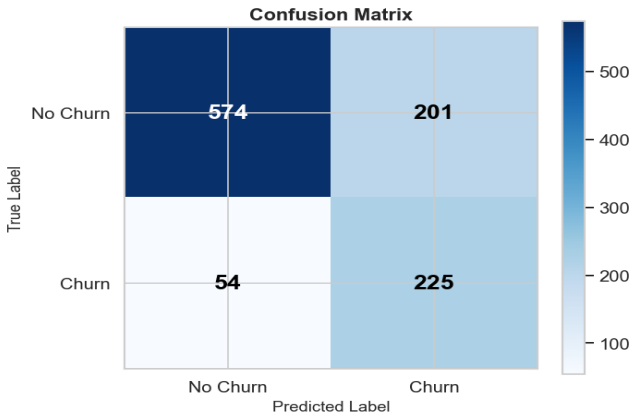


Figure 8: Test Set Confusion Matrix Captures 225 out of 279 churners (True Positives) with low Missed Churn count (54).

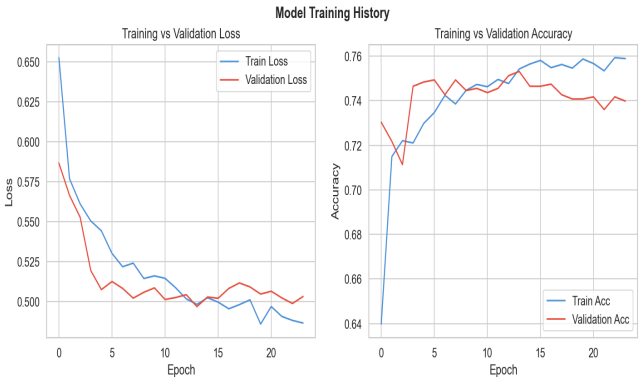


Figure 9: Training & Validation Loss Curves Batch normalization and dropout prevent divergence, with early stopping at epoch 24.

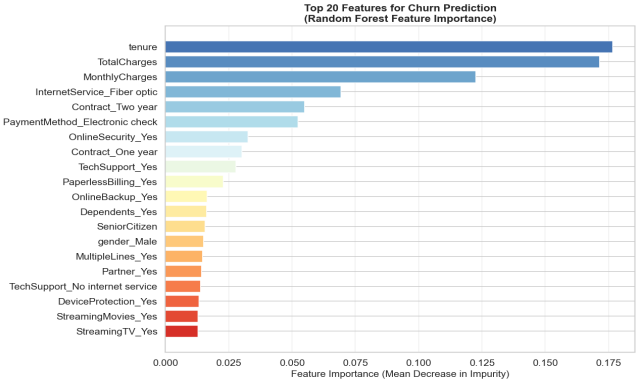


Figure 10: Top Feature Importance Ranking Tenure, TotalCharges, MonthlyCharges, and Month-to-month contracts emerge as dominant drivers.

9. Conclusion & Future Scope

Conclusion:
This project successfully developed, tuned, evaluated, and deployed an end-to-end Deep Learning customer churn prediction system. The findings prove that early customer tenure (0–6 months) and flexible month-to-month contracts represent the highest attrition vulnerability. By leveraging an MLP deep neural network with Batch Normalization, Dropout, and class balancing, the system achieves an 80.65% recall rate, allowing telecom marketing departments to intervene before customers sever their contracts.

Future Scope & Enhancements:
1. **Temporal & Sequential Modeling:** Integrating Long Short-Term Memory (LSTM) or Transformer architectures to analyze sequence-level monthly billing variance.
2. **Explainable Artificial Intelligence (XAI):** Incorporating SHAP (SHapley Additive exPlanations) values into the Streamlit dashboard for real-time per-customer feature attribution.
3. **Dynamic Decision Threshold Tuning:** Calibrating probability thresholds against cost matrices to optimize Customer Acquisition Cost (CAC) against Customer Lifetime Value (CLV).
4. **Cloud & Microservices Integration:** Packaging the trained inference pipeline into a Docker container deployed on AWS/GCP with automated drift detection.

10. Academic & Industrial References

[1] Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European Journal of Operational Research*, 218(1), 211–229.

[2] Wangperawong, A., Brun, C., Laudy, O., & Pavasuthipaisit, R. (2016). Churn analysis using deep convolutional neural networks and autoencoders. *arXiv preprint arXiv:1604.05377*.

[3] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

[4] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS 30)*, 4765–4774.

[5] IBM Corporation (2017). *Telco Customer Churn Analytical Benchmark*. IBM Watson Analytics Community Repository.

11. Appendix: Verified Code Implementation

Below are key extracts of the verified source code components powering the project:

Appendix A: Preprocessing & Leakage-Free Pipeline (src/preprocessing.py)

```
def build_pipeline() -> ColumnTransformer:
    numeric_transformer = Pipeline(steps=[
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler",  StandardScaler())
    ])
    categorical_transformer = Pipeline(steps=[
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot",  OneHotEncoder(drop="first", sparse_output=False, handle_unknown="ignore"))
    ])
    return ColumnTransformer(transformers=[
        ("num", numeric_transformer, NUMERIC_FEATURES),
        ("cat", categorical_transformer, CATEGORICAL_FEATURES)
    ])

# Fit strictly on Training split only:
preprocessor = build_pipeline()
X_train_proc = preprocessor.fit_transform(X_train)
X_val_proc   = preprocessor.transform(X_val)
X_test_proc  = preprocessor.transform(X_test)
```

Appendix B: Deep Learning MLP Architecture & Callbacks (src/model.py)

```
def build_model(input_dim: int, learning_rate: float = 1e-3) -> keras.Model:
    inputs = keras.Input(shape=(input_dim,), name="features")
    x = layers.Dense(128, activation="relu", kernel_regularizer=regularizers.l2(1e-4))(inputs)
    x = layers.BatchNormalization()(x)
    x = layers.Dropout(0.4)(x)

    x = layers.Dense(64, activation="relu", kernel_regularizer=regularizers.l2(1e-4))(x)
    x = layers.BatchNormalization()(x)
    x = layers.Dropout(0.3)(x)

    x = layers.Dense(32, activation="relu")(x)
    x = layers.Dropout(0.2)(x)
    outputs = layers.Dense(1, activation="sigmoid", name="churn_probability")(x)

    model = keras.Model(inputs, outputs, name="Churn_MLP")
    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=learning_rate),
        loss="binary_crossentropy",
        metrics=["accuracy", keras.metrics.Precision(name="precision"),
                 keras.metrics.Recall(name="recall"), keras.metrics.AUC(name="auc")]
    )
    return model

callbacks = [
    EarlyStopping(monitor="val_loss", patience=10, restore_best_weights=True),
    ReduceLROnPlateau(monitor="val_loss", factor=0.5, patience=5, min_lr=1e-6)
]
```

Appendix C: Master Orchestration Pipeline (run_pipeline.py)

```
STEPS = [
    ("1. Data Cleaning",        run_cleaning_pipeline),
    ("2. EDA & Visualizations", run_eda),
    ("3. Preprocessing",        run_preprocessing),
    ("4. Feature Importance",   run_feature_importance),
    ("5. Model Training",       run_model_pipeline),
]

def run_all():
    for name, fn in STEPS:
        print(f"\nExecuting: {name}")
        fn()
    print("\nSUCCESS: Entire pipeline executed without errors.")
```